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CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.



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Best of luck!



Question No : 1 of 52

Marks: 1 (Budgeted Time 1 Min)

The solution of $x^2 \frac{d^2 y}{dx^2} = 0$ is

Answer (Please select your correct option)

☐ $y = c_1 + c_2 x$



☐ $y = c_1 x + c_2 x^2$

☐ $y = c_1 x + c_2 x^3$

☐ None of them

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Question No : 2 of 52

Marks: 1 (Budgeted Time 1 Min)

The nature of roots of auxiliary equation deduced from Cauchy Euler equation

$$4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} + y = 0$$

Answer (Please select your correct option)

☐ Real and unequal

☐ Real and repeated

☐ Complex

☐ None of them



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Question No : 3 of 52

Marks: 1 (Budgeted Time 1 Min)

A function f is said to be convergent at a point a if it can be represented by the power series in $(x-a)$ which has

Answer (Please select your correct option)

- ☐ Positive radius of convergence
- ☐ Radius of convergence equals zero.
- ☐ None of them
- ☐ Negative radius of convergence

because in $(x-a)$
both negative and
positive value
available



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Question No : 4 of 52

Marks: 1 (Budgeted Time 1 Min)

If $E(t)=0$, $R \neq 0$ ($E(t)$ is the source voltage & R is the resistance) then electric vibration of the circuit is said to be

Answer (Please select your correct option)

☐ Free damped oscillation



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☐ Free un-damped oscillation

☐ Both damped and un-damped oscillation

☐ None of them

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Question No : 5 of 52

Marks: 1 (Budgeted Time 1 Min)

The quantity $Z = \sqrt{X^2 + R^2}$ is called

Answer (Please select your correct option)

☐ Reactance of circuit

☐ Impedance of circuit

☐ Quasi of circuit

☐ None of them



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Question No : 6 of 52

Marks: 1 (Budgeted Time 1 Min)

The time interval between two successive maxima of $x(t) = Ae^{-\lambda t} \sin[\sqrt{\omega^2 - \lambda^2} t + \phi]$ is called

Answer (Please select your correct option)

- ☐ None of them
- ☐ Both the period
- ☐ Quasi-period
- ☐ Phase period



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Question No : 7 of 52

Marks: 1 (Budgeted Time 1 Min)

The Quasi-frequency of the solution $\mathbf{x(t)}$ of free damped motion is given by the number

Answer (Please select your correct option)

☐ $\frac{\sqrt{\omega^2 - \lambda^2}}{2\pi}$



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☐ $\frac{2\pi}{\sqrt{\omega^2 - \lambda^2}}$

☐ $\sqrt{\omega^2 - \lambda^2}$

☐ None of them

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Question No : 8 of 52

Marks: 1 (Budgeted Time 1 Min)

For the equation of free damped motion $\frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = 0$ the roots are $m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}$ & $m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}$ If $\lambda^2 - \omega^2 < 0$ then system is said to be

Answer (Please select your correct option)

- ☐ Over damped
- ☐ Critically damped
- ☒ Under damped
- ☐ None of them



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Question No : 9 of 52

Marks: 1 (Budgeted Time 1 Min)

The general solution of the equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \frac{1}{25}) = 0$ is

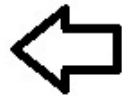
Answer (Please select your correct option)

☐ $y = c_1 J_{\frac{1}{3}}(x) + c_2 J_{-\frac{1}{3}}(x)$

☐ $y = c_1 J_{\frac{1}{4}}(x) + c_2 J_{-\frac{1}{4}}(x)$

☐ $y = c_1 J_{\frac{1}{3}}(x) + c_2 J_{-\frac{1}{3}}(x)$

☐ $y = c_1 J_{\frac{1}{25}}(x) + c_2 J_{-\frac{1}{25}}(x)$



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Question No : 10 of 52

Marks: 1 (Budgeted Time 1 Min)

$$J_{-\frac{2}{3}}(x) - J_{\frac{4}{3}}(x) =$$

Answer (Please select your correct option)

☐ $2J'_{\frac{1}{3}}(x)$

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☐ $2J'_{\frac{2}{3}}(x)$

☐ $2J'_{\frac{4}{3}}(x)$

☐ None of them

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Question No : 11 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix}$ & $B = \begin{bmatrix} x & y & z & a \\ p & q & r & b \\ l & m & n & o \end{bmatrix}$ then the order of matrix $A \times B$ is

Answer (Please select your correct option)

☐ 2×4



☐ 2×3

☐ 3×3

☐ None of them

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Question No : 12 of 52

Marks: 1 (Budgeted Time 1 Min)

The order of a matrix which contains 1 rows and m columns is

Answer (Please select your correct option)

☐ $1 \times m$



☐ $2 \times m$

☐ $m \times 1$

☐ None of them

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Question No : 13 of 52

Marks: 1 (Budgeted Time 1 Min)

Eigen value of the matrix $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$ is

Answer (Please select your correct option)

☐ $\lambda = 5, 3$

☒ $\lambda = 5, 5$

☐ $\lambda = 3, 4$

☐ None of them



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Question No : 14 of 52

Marks: 1 (Budgeted Time 1 Min)

The given system without the use of matrices $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} \sin t$ is

Answer (Please select your correct option)

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin 2t; \frac{dy}{dt} = x + y + 8 \cos 2t$

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin t; \frac{dy}{dt} = x + y + 8 \cos t$

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin t; \frac{dy}{dt} = x + y + 8 \sin t$

☐ None of them

Correct Answer



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Question No : 15 of 52

Marks: 1 (Budgeted Time 1 Min)

The given system without the use of matrices $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} e^{-t}$ is

Answer (Please select your correct option)

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin 2t; \frac{dy}{dt} = x + y + 8 \cos 2t$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^{-t}; \frac{dy}{dt} = x + y + 8e^{-t}$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^t; \frac{dy}{dt} = x + y + 8e^{-t}$

☐ None of them



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Question No : 16 of 52

Marks: 1 (Budgeted Time 1 Min)

The coefficient matrix of the following homogeneous system of differential equation $\frac{dx}{dt} = 3x + 2y$, $\frac{dy}{dt} = x + 2y$ is

Answer (Please select your correct option)

☐ $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

☐ None of them



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Question No : 17 of 52

Marks: 1 (Budgeted Time 1 Min)

The matrix $A = \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ has eigen values $\lambda = -1, -1, 5$ where $\lambda = -1$ is a

Answer (Please select your correct option)

☐ Single root of A

☐ triple root of A

☐ double root of A

☐ None of them



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Question No : 18 of 52

Marks: 1 (Budgeted Time 1 Min)

By applying the Operator method or systematic elimination on a system of linear homogeneous or linear non-homogeneous differential equations we always get a

Answer (Please select your correct option)

☐ Single linear differential equation



☐ Double linear differential equation

☐ Partial linear differential equation

☐ None of them

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Question No : 19 of 52

Marks: 1 (Budgeted Time 1 Min)

If L denote the linear differential operators with constant coefficients, then $L_1 L_4 - L_2 L_3$ represents the

Answer (Please select your correct option)

☐ $\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$

☐ $\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$

☐ $\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$

☐ None of them



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Question No : 20 of 52

Marks: 1 (Budgeted Time 1 Min)

The matrix $\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ is

Answer (Please select your correct option)

☐ Singular matrix



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☐ Non singular matrix

☐ Diagonal matrix

☐ Scalar Matrix

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Question No : 21 of 52

Marks: 1 (Budgeted Time 1 Min)

The Differential Equation $(x^2 - 4)y'' - 10xy' + y = 0$ has singularity at

Answer (Please select your correct option)

☐ $x = \pm 1$

☐ $x = \pm 2$

☐ $x = \pm 3$

☐ $x = \pm 4$



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Question No : 22 of 52

Marks: 1 (Budgeted Time 1 Min)

Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systematic elimination of the

Answer (Please select your correct option)

☐ Dependent variables

☐ Independent variable

☐ Choice variable

☐ None of them



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Question No : 23 of 52

Marks: 1 (Budgeted Time 1 Min)

The non-zero solution of the system exists only when

Answer (Please select your correct option)

☐ $\det(A - \lambda I) = 1$

☒ $\det(A - \lambda I) = 0$

☐ $\det(A - \lambda I) = -1$

☐ $\det(A - \lambda I) \neq 0$



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Question No : 24 of 52

Marks: 1 (Budgeted Time 1 Min)

The solution of the linear first order differential equation $\frac{dy}{dx} - 2y = 0$ is

Answer (Please select your correct option)

☐ $y = e^{2x+c}$



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☐ $y = \sum_{n=0}^{\infty} \frac{x^n}{4n!}$

☐ Both 1) and 2)

☐ None of them

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Question No : 25 of 52

Marks: 1 (Budgeted Time 1 Min)

Ordinary points of $(x^2 - 64)(x^2 - 36)y'' + xy' - y = 0$ are

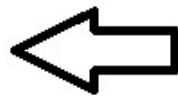
Answer (Please select your correct option)

☐ 0,1

☐ 8,-8

☐ 6,-6

☐ None of others.



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Question No : 26 of 52

Marks: 1 (Budgeted Time 1 Min)

Ir-regular singular point of the equation $(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$ is

Answer (Please select your correct option)

☐ $x = 2$

☐ $x = -2$

☐ $x = -2, 2$

☐ None of them



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Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

The matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has

Answer (Please select your correct option)

☐ Real and unequal value

☐ Repeated & real eigen value

☐ Complex eigen value

☐ None of them



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Question No : 28 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, then eigen values are

Answer (Please select your correct option)

☐ 1,2

☐ 0,1

☐ 0,2

☐ None of them



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Question No : 29 of 52

Marks: 1 (Budgeted Time 1 Min)

Let λ be an eigen value of a non zero square matrix A. Then the equation $\det(A - \lambda I) = 0$ is called

Answer (Please select your correct option)

☐ Trivial equation

☐ Characteristics equation

☐ Non-trivial equation

☐ None of them



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Question No : 30 of 52

Marks: 1 (Budgeted Time 1 Min)

Eigen values of the following homogeneous system of Differential equation $\frac{dx}{dt} = x$, $\frac{dy}{dt} = 2x + 2y$ with coefficient matrix $\begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

Answer (Please select your correct option)

☐ $\lambda = 1, 2$



☐ $\lambda = 2, 2$

☐ $\lambda = 1, 1$

☐ None of them

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Question No : 31 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the following may not be considered as integration technique

Answer (Please select your correct option)

- ☐ By Parts
- ☐ By substitutions
- ☐ By Partial Fractions
- ☐ None of these



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Question No : 32 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the following equations satisfy the differential equation $\frac{dy}{dx} = x$

Answer (Please select your correct option)

☐ $2y = x^2 + c$



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☐ $y = x^2 + c$

☐ $y = x^2$

☐ $y = x + c$

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Question No : 33 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation $(3x^2y + 2) dx + (x^3 + y) dy = 0$ is -----.

Answer (Please select your correct option)

☐ Exact



☐ Non-exact

☐ Separable

☐ Homogenous

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Question No : 34 of 52

Marks: 1 (Budgeted Time 1 Min)

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

☐ $v = y^{n-1}$

☐ $v = y^{1-n}$

☐ $v = y^n$

☐ $v = y'$



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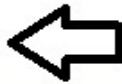
Question No : 35 of 52

Marks: 1 (Budgeted Time 1 Min)

A differential equation of the form $\frac{dy}{dx} = f(x, y)$ is said to be homogeneous if $f(tx, ty) = \dots$.

Answer (Please select your correct option)

☐ $f(x, y)$



☐ $f(x)$

☐ $f(y)$

☐ C

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Question No : 36 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation ----- is separable.

Answer (Please select your correct option)

☐ $x(x+y) \frac{dy}{dx} = 4$

☐ $\frac{dy}{dx} = \frac{x^2}{x+xy}$

☐ $\frac{dy}{dx} = \frac{y}{1+xy^3}$

☐ $\frac{dy}{dx} = \frac{xy+3}{1+2xy}$



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Question No : 37 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation ----- is not separable.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = \frac{2xy+3}{3+4xy^2}$



☐ $\frac{dy}{dx} = \frac{x+1}{x+xy^2}$

☐ $\frac{dy}{dx} = \frac{1}{x^2y+4y}$

☐ $\frac{dy}{dx} = 1+y+x+xy$

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Question No : 38 of 52

Marks: 1 (Budgeted Time 1 Min)

If the equation $M(x, y)dx + N(x, y)dy = 0$ is not exact and $\mu = \frac{N_x - M_y}{M}$ is a function of y only, then the integrating factor is given by-----.

Answer (Please select your correct option)

☐ $I.F = e^{\int \mu dy}$

☐ $I.F = e^{\int 2\mu dy}$

☐ $I.F = e^{\int \frac{1}{2}\mu dy}$

☐ $I.F = e^{-\int \mu dy}$



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Question No : 39 of 52

Marks: 1 (Budgeted Time 1 Min)

For the solution of the equation $\frac{-1}{y-1} = x + c$ with $y(0) = 3$, the value of c is ---.

Answer (Please select your correct option)

☐ 0

☐ 1

☐ -1/2

☐ -1



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Question No : 40 of 52

Marks: 1 (Budgeted Time 1 Min)

If $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$ is the solution of $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = 0$, then which of the following is the most accurate option for $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = e^{2x} \sin 3x$?

Answer (Please select your correct option)

- ☐ Its general form of the particular solution will be $Ae^{2x} + B \sin x + C \cos x$.
- ☐ Its general form of the particular solution will be $e^{2x}(A \sin x + B \cos x)$.
- ☐ Its general form of the particular solution will be $e^{2x}(Ax \sin x + Bx \cos x)$.
- ☐ Its general form of the particular solution will be $e^{2x}(Ax \sin 3x + Bx \cos 3x)$.



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Find the eigenvalues of the following system

$$X' = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix} X$$

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Answer (Please [click here](#) to Add Answer)

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$$\begin{pmatrix} 3-a & -9 \\ 4 & -3-a \end{pmatrix}$$

$$-9 - 3a + 3a + a^2 + 36 = 0$$

$$a^2 + 27$$

hare use the lamanda..! not a

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Is generally in matrices following laws hold or not?

- 1- Associative Law
- 2- Distributive Law
- 3- Commutative Law

Answer (Please [click here](#) to Add Answer)

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yes, $A + B + C = A + (B + C) = (A + B) + C$ (Associative law of addition)

$A + B = B + A$ (Commutative law of addition)

$A(B + C) = AB + AC$ (Distributive law)

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Define regular and irregular singular points?

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Answer (Please [click here](#) to Add Answer)

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Definition: Regular and Irregular Singular Points

A Singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a *regular singular point* if both $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0 . A singular point that is not regular is said to be an *irregular singular point* of the equation.

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Question No : 44 of 52

Marks: 2 (Budgeted Time 4 Min)

Solve the differential equation:

$$\frac{dy}{dx} = \frac{x^2}{2y}$$

Answer (Please [click here](#) to Add Answer)

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$$2 \int y dy = \int x^2 dx$$

$$y^2 = \frac{x^3}{3} + c$$

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Write down the procedure of solution of the system of differential equations by "Operator Method".

Answer (Please [click here](#) to Add Answer)

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Systematic Elimination: Operator Method

- ☐ This method of solution of a system of linear homogeneous or linear nonhomogeneous differential equations is based on the process of systematic elimination of the dependent variables.
 - ☐ This elimination provides us a single differential equation in one of the dependent variables that has not been eliminated.
- >> This equation would be a linear homogeneous or a linear non-homogeneous differential equation and can be solved by employing one of the methods discussed earlier to obtain the dependent variables.

Notice that the analogue of multiplying an algebraic equation by a constant is operating on a differential equation with some combination of derivatives.

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Question No : 49 of 52

Marks: 5 (Budgeted Time 10 Min)

Find the general solution of the given differential equation on $(0, \infty)$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \frac{1}{64}\right)y = 0$$

Answer (Please [click here](#) to Add Answer)

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Question No : 50 of 52

Marks: 5 (Budgeted Time 10 Min)

Write the following system in matrix form

$$\frac{dx}{dt} = x - y + z + t - 1$$

$$\frac{dy}{dt} = 2x + y - z - 3t^2$$

dz

Answer (Please [click here](#) to Add Answer)

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What is indicial equation and exponent in the differential equation $xy'' + 3y' - y = 0$ with $x=0$ regular singular point?

Answer (Please [click here](#) to Add Answer)



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Determine the order and state the linearity of each of the following differential equations.

1) $\left(\frac{d^3 y}{dx^3}\right)^4 + 2 \frac{dy}{dx} = \sin x$

2) $\frac{dy}{dx} - 2xy = x^2 - x$

Answer (Please [click here](#) to Add Answer)

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Answers to Above Exercises

1. order 3 , non linear.

2. order 1 , linear.

3. order 1 , non linear.

4. order 2 , linear.

1. $\left(\frac{d^3 y}{dx^3}\right)^4 + 2 \frac{dy}{dx} = \sin x$

2. $\frac{dy}{dx} - 2xy = x^2 - x$

3. $\frac{dy}{dx} - \sin y = -x$

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